# JAD **KHAYYATI**

Software Engineer

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## SUMMARY

Computer Science student at Arizona State University, graduating August 2027. Building an AI-powered maintenance platform with Honeywell Aerospace in Python, FastAPI and AWS, and built and launched a full staffing platform on my own for a paying client using Next.js, Supabase and Twilio. At school I've worked deep in the stack, from Linux kernel modules in C to a compiler front-end in C++.

## SKILLS

- Languages: Python, Java, C, C++, C#, JavaScript, TypeScript, Structured Query Language (SQL)
- Backend and cloud: FastAPI, REST APIs, Amazon Web Services (AWS: S3, SageMaker, IAM), role-based access control (RBAC)
- Web: Next.js, React, HTML5, CSS, Supabase (PostgreSQL), Twilio, ASP.NET
- Tools: Git/GitHub, Docker, Linux, Jira, Scrum, pytest, JUnit, Unified Modeling Language (UML)
- Systems programming: Linux kernel modules, multithreading, compilers
- API integration
- Responsive design
- Soft skills: client communication, independent end-to-end delivery, teamwork, learning new tools quickly

## EDUCATION

**Expected 08/2027**

**Bachelor of Science (B.S.): Computer Science**
**Arizona State University (ASU Online)** - Tempe, Arizona

- Transferred from Michigan State University, East Lansing, Michigan
- Relevant coursework: Operating Systems, Principles of Programming Languages, Software Analysis and Design, Distributed Software Development, Database Management, Introduction to Artificial Intelligence (AI), Theory of Computation
- GPA: 3.21 / 4.00

## INDUSTRY PROJECT

**09/2026 - Current**

**Honeywell Aerospace: AeroSecure AI**
**Industry capstone with Honeywell Aerospace, Arizona State University** - Python, FastAPI, scikit-learn, pandas, AWS (S3, SageMaker, IAM), pytest, Git/GitHub, Jira, Scrum

*AI-powered aerospace maintenance platform that predicts how long replacement parts will take to arrive, recommends which supplier to order from, and controls who can access restricted data. Team of 4 in 2-week Scrum sprints, with weekly reviews from Honeywell engineers.*

- Built role-based access control (RBAC) for 20,000 order records (student / mentor / admin) with deny-by-default rules and an audit log of every request, exposed through a FastAPI endpoint
- Built a supplier recommendation engine that combines the machine learning model's delivery predictions with shipment restriction rules to recommend a supplier, escalate, or block an order, and added it to the backend as a REST endpoint
- Wrote automated tests for my features with pytest and unittest, validated against Honeywell's official policy and maintenance scenarios (all passing)
- Configured Amazon Web Services (AWS) S3, IAM roles and SageMaker for cloud model training, and working on enforcing data access tiers in S3
- Team result: our delivery-time model (scikit-learn random forest) beats the supplier quote baseline, 5.2 vs 6.6 days average error

## EXPERIENCE

**06/2026 - Current**

**Full-Stack Developer (Contract)**
**Cocon Éducation** - Quebec, Canada (remote)

- Sole developer on a substitute-teacher staffing platform for Quebec childcare centres: designed, built and launched it from scratch with Next.js, Supabase and Twilio
- Built three separate portals for admins, centres and substitutes, each with its own views and permissions
- Added Short Message Service (SMS) notifications between centres and substitute teachers using the Twilio API
- Negotiated the contract directly with the client and launched the platform, which now gets around 200-300 visits a day

**05/2026 - Current**

**Freelance Web Developer**
**Self-employed** - Morocco

- Built a real estate website for Youssra Terre SARL in vanilla HTML, CSS and JavaScript
- Building a French-language portfolio site for Green Architects, handling hosting, domain and content collection for a non-technical client

## PROJECTS

**Linux Kernel Modules**
**Operating Systems (CSE 330)** - C, Linux

- Wrote kernel modules in C: a producer–consumer system with kernel threads and synchronization, a zombie-process killer, a memory allocation module, and a block-device I/O module controlled through input/output control (IOCTL) calls
- Built and tested every module in a sandboxed virtual machine on pwn.college

**Compiler Front-End**
**Principles of Programming Languages (CSE 340)** - C++, Docker

- Implemented a recursive descent parser, grammar analysis and intermediate representation (IR) generation for a small language in C++

## LANGUAGES

**English:** Proficient (C2)
**Arabic:** Proficient (C2)
**French:** Native
**Spanish:** Intermediate (B1)